

Matthew Minwoo Kim

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Education

Seoul National University <i>B.S. in Computer Science and Engineering</i> <i>Interdisciplinary Major in AI — Kim Jaechul AI Class</i> - GPA: 4.21/4.3 (76 credits)	Mar. 2023 – Present Seoul, South Korea
Korea Science Academy of KAIST <i>High School Diploma, Computer Science</i> - GPA: 4.14/4.3 (top 10% of the class)	Mar. 2020 – Feb. 2023 Busan, South Korea
University of Cambridge <i>International Summer School - Business and Entrepreneurship</i>	Summer 2022 Cambridge, United Kingdom

Research Experience

Vision and Learning Lab , Seoul National University <i>Undergraduate Research Intern (Advisor: Prof. Gunhee Kim)</i> Conducting research with interest of multimodal AI.	Apr. 2026 – Present
Intelligent Data Systems Lab , Seoul National University <i>Undergraduate Research Opportunity Program (Advisor: Prof. Sang-goo Lee)</i> Researched with topic of “Text-to-SQL Generation using Intermediate Representation” and “Resolving Ambiguity in Text-to-SQL Generation”	Jul. 2024 – Mar. 2025
Computer Science Research Group , Korea Science Academy of KAIST <i>Student Researcher</i> - Researched with the topic of “A Secret Sharing Scheme without Third-Party Dealers” - Resulted in Regeneron ISEF South Korea Finalist	Feb. 2021 – Feb. 2022

Professional Experience

Republic of Korea Air Force <i>AI Researcher</i> Conducted research and development of military-specific Artificial Intelligence systems, specializing in Computer Vision and Speech.	Oct. 2024 – Jul. 2026
AIBIZON AI Lab <i>Research Team Intern</i> - Conducted an empirical study of RAPTOR(Recursive Abstractive Processing for Tree-Organized Retrieval) - Engineered a CRDT-based (Conflict-free Replicated Data Type) program to enable synchronous connections and collaborative state management on Slack.	Mar. 2024 – Apr. 2024
Hyperithm <i>Research Team & Infra Team Intern</i> - Developed a high-frequency trading (HFT) optimized cryptocurrency exchange data feed to ensure ultra-low latency market data ingestion. - Developed and maintained quantitative trading strategy tools to support internal research and trading operations.	Jun. 2023 – Aug. 2023

Awards & Achievements

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| Army Chief of Staff Award , Military AI Contest (MAICON) | 2025 |
| ◦ 2nd place, national-level AI competition | |
| South Korea Finalist , Regeneron ISEF (International Science and Engineering Fair) | 2022 |
| ◦ Systems Software category (Team Leader) | |
| Grand Prize , R&E Project Presentation Contest for the Scientific Talent | 2021 |
| ◦ Ministry of Science and ICT Award | |

Publications

- [1] Seokhun Ji, Euisuh Jeong, **Minwoo Kim**, Hyojin Lee, "Deep Learning for Pavement Management System: Proposing an Automated Pipeline for Pavement Condition Index (PCI) Assessment", in *Journal of the Korean Society of Airport*, 1(1), 52-65.
- [2] Euisuh Jeong, Seokhun Ji, **Minwoo Kim**, Hyojin Lee, "ROKAF Runway Crack Dataset: Construction and Application of a Large-Scale AI-Based Runway Defect Detection Dataset", in *Journal of the Korean Society of Airport*, 1(2), 114-131.

Certificates

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| SQL Developer | Jun 2024 |
| Craftsman Information Processing | Apr 2024 |

Skills

- **Languages:** Korean (Native), English (Advanced)
- **Programming Languages:** Python (PyTorch, TensorFlow), C++, Java, Typescript, Rust

Additional Information

- **Citizenship:** United States & South Korea
- **Military Service:** Serving as AI Researcher in ROK Air Force (Oct.2024–Jul.2026).

Research Interests

- Multimodal AI
- Representation Learning
- Algebra